Noise-Aware Spectrally Adaptive Wiener Deconvolution

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***Abstract*—Deconvolution can recover resolution lost to an optical point spread function (PSF), but it amplifies noise at exactly the frequencies it tries to restore. Regularized inverse filters control this with a parameter λ that is usually hand-tuned or chosen per image by generalized cross-validation (GCV).**

We propose a spectrally adaptive Wiener filter whose frequency-dependent regularizer $\lambda(\omega)$ is estimated entirely from the blurred image. The noise power comes from the frequency band where the optical transfer function is negligible, and the signal power from a power-law fit to the noise-corrected, de-blurred spectrum. The method has no tunable regularization parameter and costs a few FFTs per image.

On a controlled benchmark built from fluorescence-microscopy images (FMD dataset; 8 test crops, 3 Gaussian PSF widths, 4 noise levels, 5 noise seeds), the method:

- **comes within 0.15–0.39 dB PSNR of the oracle Wiener filter, which uses the true signal spectrum;**
- **has median paired PSNR gains of 0.716 dB over constant-NSR Wiener, 3.499 dB over Richardson–Lucy, and 3.110 dB over unsupervised Wiener;**
- **is effectively tied with Tikhonov–GCV (median difference 0.004 dB).**

The reported benchmark uses eight 256×256 test crops, three Gaussian PSF widths, four additive Gaussian noise levels, and five noise seeds per condition. The updated aggregate tables and paired comparisons are used throughout this draft; real-image findings are treated as noise-amplification and resolution evidence rather than ground-truth reconstruction accuracy.

I. Introduction

Every optical imaging system blurs. In fluorescence microscopy and optical coherence tomography (OCT), a point in the sample is recorded as a spot whose shape is the system's point spread function (PSF), limited by diffraction and by the optics. The recorded image is therefore the true structure convolved with the PSF, plus detector noise: photon (Poisson) noise and read-out (Gaussian) noise in microscopy, and speckle in OCT.

Deconvolution inverts this convolution. In the frequency domain, blur multiplies each spatial frequency by the optical transfer function $H(\omega)$, which falls towards zero at high frequencies. Undoing the blur means dividing by $H(\omega)$. Where $H(\omega)$ is small, that division amplifies noise far more than it restores signal. Every practical deconvolution method is therefore a regularized inverse: it trades resolution against noise amplification through a regularization parameter λ. How λ is chosen decides the result. Common practice is to

- tune one constant by eye or on example images;
- pick it per image with a criterion such as generalized cross-validation (GCV) [Golub et al.];
- run an iterative method such as Richardson–Lucy and stop it early, so the iteration count acts as λ.

In each case λ is a single number applied to all frequencies. The statistically optimal (Wiener) solution instead uses a frequency-dependent $\lambda(\omega) = S_n(\omega)/S_x(\omega)$, the ratio of noise power to signal power. That requires the signal spectrum S_x, which is unknown for a real image.

The deconvolution literature for OCT [Liu et al. 2009; Woolliams et al. 2010; Schmitt 1998] and blind deconvolution [Moiseev et al. 2011] shows the gains available and also the sensitivity of these methods to noise and PSF errors.

Contributions.

1. A noise-adaptive regularizer $\lambda(\omega) = \hat{\sigma}^2/(A\,|\omega|^{-\beta})$. Here $\hat{\sigma}^2$ is estimated from the band where $|H(\omega)|^2 < \varepsilon$, and (A, β) from a power-law fit to the noise-corrected, de-blurred radial spectrum. No regularization parameter is tuned.

2. A controlled benchmark with exact ground truth, built from real fluorescence-microscopy images. It covers 3 PSF widths, 4 Gaussian noise levels and 3 Poisson–Gaussian photon levels, with 5 noise seeds each. It compares six methods, including an oracle Wiener upper bound, on fidelity (PSNR, SSIM, log-spectral distance), resolution (edge FWHM, MTF50), ringing, noise amplification and runtime.

3. An evaluation on real noise. The 50-frame averages of the FMD dataset serve as a clean reference, so we can measure the noise amplification of each filter on real Poisson–Gaussian noise. We also give a qualitative OCT demonstration.

4. An ablation and robustness study: oracle noise, oracle spectrum and constant λ variants, PSF-width mismatch of $\pm 30\,\%$, and boundary handling.

The rest of the report is organized as follows. Sections 2–4 give the imaging model, the regularized inverse and the proposed method. Sections 5–6 describe the baselines and the experimental setup. Sections 7–10 present the results. Section 11 discusses limitations.

II. Forward Imaging Model

A. Continuous-to-discrete model

We model a grayscale image as a 2-D discrete signal on an $M \times N$ pixel grid, with intensities normalized to $[0, 1]$ (Section 6.3). The observed image y is the unknown object x convolved with the PSF h, plus additive noise n:

$$\begin{aligned} y[m,n] &= (h * x)[m,n] + n[m,n] \\ &= \sum_{k,l} h[k,l]\, x[m-k,\, n-l] + n[m,n]. \end{aligned} \tag{1}$$

Table 1: Notation and conventions used in the imaging model.

Symbol	Meaning	Units / convention
$x[m, n]$	true object intensity	normalized, 0–1
$h[k, l]$	point spread function, odd size $K \times K$, centred, $\sum h = 1$	dimensionless; $\sum h = 1$ preserves mean brightness
$n[m, n]$	noise	same units as x
$y[m, n]$	recorded image	same units as x
m, n	pixel indices	samples
$\omega = (\omega_x, \omega_y)$	spatial frequency	cycles/pixel, each component in $[-0.5, 0.5)$
$\lvert\omega\rvert$	radial frequency	cycles/pixel, $0 \ldots 0.707$

Table 1 summarizes the notation used throughout. Convolution in the experiments is circular (periodic boundaries) unless stated otherwise. That makes the discrete Fourier transform (DFT) diagonalize the blur exactly:

$$Y(\omega) = H(\omega)\, X(\omega) + N(\omega), \tag{2}$$

where capitals denote 2-D DFTs over the $M \times N$ grid. $H(\omega)$ is the optical transfer function (OTF), and $|H(\omega)|$ is the modulation transfer function (MTF). To compute H without a phase ramp, the $K \times K$ PSF is zero-padded to $M \times N$ and circularly shifted so that its centre sits at pixel $(0, 0)$ (psf2otf in lib/forward.py). Section 9.2 studies the more realistic mirror ("reflect") boundary.

B. PSF models

Gaussian PSF. This is the main model throughout:

$$h[k, l] \propto \exp\left(-\frac{k^2 + l^2}{2\sigma_h^2}\right), \tag{3}$$

for $|k|, |l| \le 4\sigma_h$, truncated to a square of side $8\sigma_h + 1$ (rounded to odd) and normalized to unit sum. We use $\sigma_h \in \{1.5, 2.5, 4.0\}$ pixels. A Gaussian has a Gaussian transfer function:

$$H(\omega) = \exp(-2\pi^2 \sigma_h^2\, |\omega|^2)\,, \tag{4}$$

so the MTF of the blur alone falls to 0.5 at $|\omega| = \sqrt{\ln 2/(2\pi^2)}\,/\,\sigma_h \approx 0.187/\sigma_h$ cycles/pixel: 0.125, 0.075 and 0.047 for our three widths. At $|\omega| = 0.5$ the transfer is about 1.5×10^{-5} for $\sigma_h = 1.5$ and about 10^{-34} for $\sigma_h = 4$. Those frequencies are effectively lost and can only be recovered by amplifying noise (Fig. 1, right panel).

The width relates to the more familiar full width at half maximum by FWHM $= 2\sqrt{2 \ln 2}\,\sigma_h \approx 2.355\,\sigma_h$, giving 3.5, 5.9 and 9.4 pixels.

C. Noise models

1. Additive white Gaussian noise, $n \sim \mathcal{N}(0, \sigma_n^2)$ independently per pixel, with $\sigma_n \in \{0.005, 0.01, 0.02, 0.05\}$. Its power spectrum is flat. With the unnormalized DFT, $\mathbb{E}|N(\omega)|^2 = MN\,\sigma_n^2$ at every ω. We therefore write all power spectra per pixel, $S(\omega) = \mathbb{E}|\cdot(\omega)|^2/(MN)$, so the noise power spectral density is simply $S_n(\omega) = \sigma_n^2$.

2. Poisson–Gaussian noise, the standard model for fluorescence detectors and the noise in the FMD dataset [Zhang et al. 2019]:

$$y = \frac{1}{p}\,\text{Poisson}\big(p\,(h * x)\big) + n_r, \tag{5}$$

with $n_r \sim \mathcal{N}(0, \sigma_r^2)$, where p is the photon count at intensity 1.0 ($p \in \{30, 100, 300\}$; larger p means less noise) and $\sigma_r = 0.01$ is the read-noise level. The noise variance depends on the signal,

$$\text{Var}[y] = \frac{h * x}{p} + \sigma_r^2. \tag{6}$$

The linear methods in this report treat it through its spatial average,

$$\sigma_{\text{eff}}^2 = \frac{\text{mean}(h * x)}{p} + \sigma_r^2. \tag{7}$$

3. Speckle (OCT) is multiplicative, $y = x \cdot s$. Taking the logarithm turns it into an additive term,

$$\log y = \log x + \log s, \tag{8}$$

so the OCT experiments (Section 10.2) operate on log intensity. Log-speckle is neither Gaussian nor white in general. This is a limitation of the OCT demonstration (Section 11).

D. Assumptions

- Shift-invariant blur. One PSF applies to the whole field of view. Real objectives vary across the field and with depth.
- Known PSF. In the synthetic experiments the true PSF is used for deconvolution, except in the mismatch study (Section 9.1). On real images it is assumed to be Gaussian, with σ_h swept over $\{1, 1.5, 2\}$ px (Section 10).
- Noise independent of the object (Gaussian case) or Poisson–Gaussian with known structure but unknown level (microscopy case). None of the methods is given the noise level, except the oracle.
- 2-D processing. Each image is processed independently. There is no 3-D (z-stack) deconvolution.

Figure 1 (figures/fig1_forward_model.png): clean development crop, Gaussian PSF ($\sigma_h = 2.5$ px), blurred and noisy observation ($\sigma_n = 0.02$), and $|H|$ along one frequency axis for the three PSF widths. A dotted line marks $|H| = \sqrt{\varepsilon}$, the start of the noise-only band used in Section 4.

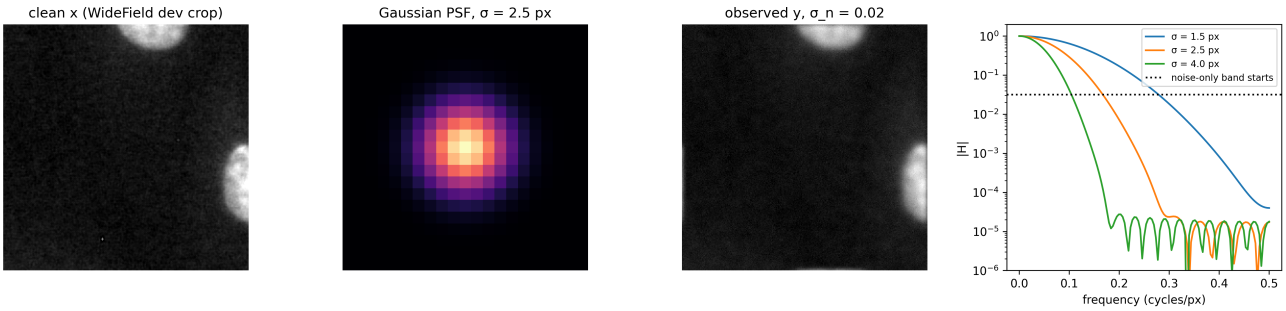


Fig. 1. Forward imaging model: a clean fluorescence-microscopy crop, Gaussian PSF, blurred/noisy observation, and the magnitude of the OTF for the three PSF widths.

III. Regularized Inverse Problem

A. The inverse filter and its noise gain

Dividing the observation by the transfer function gives the inverse filter:

$$\hat{X}_{\text{inv}}(\omega) = \frac{Y(\omega)}{H(\omega)} = X(\omega) + \frac{N(\omega)}{H(\omega)}. \tag{9}$$

The first term is perfect recovery. The second is the noise divided by H. For white noise of variance σ_n^2, the variance of the restoration error is, by Parseval:

$$\frac{\mathbb{E}\,\|\hat{x}_{\text{inv}} - x\|^2}{MN} = \sigma_n^2 \cdot \frac{1}{MN} \sum_{\omega} \frac{1}{|H(\omega)|^2}. \tag{10}$$

For a Gaussian PSF, $|H|^{-2}$ grows like $\exp(4\pi^2\sigma_h^2|\omega|^2)$. At $\sigma_h = 1.5$ and $|\omega| = 0.7$ this factor is about 10^{19}. That is why the inverse filter is unusable.

Our implementation confirms it. On a noiseless blurred image ($\sigma_h = 1.5$) the inverse filter recovers x to a maximum error of 2.8×10^{-8}, which verifies the OTF construction. Adding noise of only $\sigma_n = 0.001$ raises the maximum error to 1.4×10^5. On the full benchmark the inverse filter scores 3.4 dB PSNR (Section 7).

B. Tikhonov regularization

Tikhonov regularization chooses the image that fits the data while staying smooth:

$$\hat{x} = \arg\min_x \|y - h * x\|^2 + \lambda \|\ell * x\|^2, \tag{11}$$

where ℓ is the discrete Laplacian

$$\ell = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}. \tag{12}$$

Both terms are diagonal in the Fourier domain, so the minimizer is found frequency by frequency. Setting the derivative with respect to $X(\omega)^*$ to zero gives:

$$\hat{X}(\omega) = \frac{H^*(\omega)\, Y(\omega)}{|H(\omega)|^2 + \lambda\, |L(\omega)|^2}. \tag{13}$$

The scalar λ is chosen by generalized cross-validation [Golub, Heath & Wahba 1979]. Writing the filter gain as

$$\varphi(\omega) = \frac{|H|^2}{|H|^2 + \lambda |L|^2}, \tag{14}$$

the GCV score is:

$$\text{GCV}(\lambda) = \frac{\sum_\omega |1 - \varphi(\omega)|^2\, |Y(\omega)|^2}{\left(\sum_\omega (1 - \varphi(\omega))\right)^2}, \tag{15}$$

the residual energy divided by the squared effective number of degrees of freedom left in the residual. We minimize it over 61 log-spaced values $\lambda \in [10^{-6}, 1]$. GCV needs no knowledge of x or σ_n.

C. The Wiener filter as the optimal frequency-dependent λ

Suppose x and n are uncorrelated, zero-mean, wide-sense-stationary random fields with power spectra $S_x(\omega)$ and $S_n(\omega)$. We look for a linear filter $G(\omega)$, so that $\hat{X} = G \cdot Y$, that minimizes the mean squared error $\mathbb{E}|X - GY|^2$ at every ω. Expanding with $Y = HX + N$:

$$\mathbb{E}|X - GY|^2 = |1 - GH|^2 S_x + |G|^2 S_n. \tag{16}$$

Differentiating with respect to G^* and setting the result to zero gives the Wiener filter:

$$\begin{aligned} G_W(\omega) &= \frac{H^*(\omega)\, S_x(\omega)}{|H(\omega)|^2 S_x(\omega) + S_n(\omega)} \\ &= \frac{H^*(\omega)}{|H(\omega)|^2 + \lambda(\omega)}, \end{aligned} \tag{17}$$

$$\lambda(\omega) = \frac{S_n(\omega)}{S_x(\omega)}. \tag{18}$$

This is the regularized inverse from the project brief, $\hat{X} = H^*Y/(|H|^2 + \lambda(\omega))$. It identifies $\lambda(\omega)$ as the noise-to-signal ratio (NSR) at each frequency. The three practical choices differ only in how they set λ (Table 2):

Tikhonov with a Laplacian is itself a Wiener filter with the implicit signal model $S_x \propto |\omega|^{-4}$, since $|L(\omega)|^2 \approx (2\pi|\omega|)^4$ at low frequency. Its single λ sets the ratio of noise to that fixed spectral shape. The proposed method estimates both the noise level and the spectral exponent β from the data.

Table 2: How each method sets the regularization $\lambda(\omega)$, and what it needs as input.

Method	$\lambda(\omega)$	Needs
Inverse	0 (here 10^{-30}, a division guard)	nothing; unusable with noise
Constant-NSR Wiener	one constant, tuned on development images	a tuning set
Tikhonov–GCV	$\lambda_{\mathrm{GCV}}\,\lvert L(\omega)\rvert^2$, which grows as $\lvert\omega\rvert^4$	nothing (λ from y)
Oracle Wiener	$\sigma_n^2/S_x(\omega)$ from the true x	ground truth; upper bound only
Proposed	$\hat{\sigma}^2/(A\,\lvert\omega\rvert^{-\beta})$, which grows as $\lvert\omega\rvert^{\beta}$	nothing ($\hat{\sigma}$, A, β from y)

D. Noise gain and resolution of a linear restoration

For any linear filter G, white input noise of variance σ_n^2 leaves with variance $\sigma_n^2 \cdot \mathrm{mean}_\omega |G(\omega)|^2$. The noise gain is therefore:

$$g_n = \sqrt{\frac{1}{MN}\sum_{\omega} |G(\omega)|^2}. \tag{19}$$

The resolution delivered is set by the transfer function of the whole chain, blur followed by restoration:

$$\begin{aligned} T(\omega) &= G(\omega)H(\omega) \\ &= \frac{|H(\omega)|^2}{|H(\omega)|^2 + \lambda(\omega)} \in [0,1]. \end{aligned} \tag{20}$$

We summarize it by its MTF50, the radial frequency where T first drops below 0.5. Every regularized inverse is a point on a trade-off: smaller λ raises the MTF50 but also raises g_n. Section 10 measures both on real images.

IV. Proposed Method: Spectrally Adaptive Wiener Deconvolution

The Wiener filter (Section 3.3) is optimal, but it needs $S_n(\omega)$ and $S_x(\omega)$. We estimate both from the single blurred, noisy image y, with no hand-tuned regularization parameter.

A. Noise level from the noise-only band

For white noise, $S_n(\omega) = \sigma_n^2$ at every frequency. Wherever the blur has removed the signal, the observed spectrum is noise alone. Define the noise-only band:

$$\mathcal{B} = \{\omega : |H(\omega)|^2 < \varepsilon\}, \qquad \varepsilon = 10^{-3}. \tag{21}$$

On $\mathcal{B}$, $\mathbb{E}|Y(\omega)|^2/(MN) = |H|^2 S_x + \sigma_n^2 \approx \sigma_n^2$, with a relative bias of at most $\varepsilon S_x/\sigma_n^2$. The estimate is the band average:

$$\hat{\sigma}^2 = \frac{1}{|\mathcal{B}|}\sum_{\omega\in\mathcal{B}} \frac{|Y(\omega)|^2}{MN}. \tag{22}$$

$\mathcal{B}$ typically holds tens of thousands of frequency bins, so the estimate has low variance. For a Gaussian PSF the band starts at $|\omega| = \sqrt{\ln(1/\varepsilon)}/(2\pi\sigma_h)$: 0.28, 0.17 and 0.10 cycles/pixel for $\sigma_h = 1.5, 2.5, 4$. If the PSF is so narrow that $\mathcal{B}$ has fewer than 50 bins, the method falls back to the wavelet median-absolute-deviation estimator [Donoho & Johnstone 1994] (skimage.restoration.estimate_sigma). ε was fixed on the development set and is stored in configs/params.yaml.

B. Signal spectrum from a power-law fit

Natural and biological images have power spectra that fall approximately as a power of radial frequency [Field 1987; van der Schaaf & van Hateren 1996]:

$$S_x(\omega) \approx A\,|\omega|^{-\beta}. \tag{23}$$

We estimate A and β as follows.

1. Compute the per-pixel periodogram of the mean-removed image, $P(\omega) = |Y(\omega)|^2/(MN)$.
2. Group frequencies into 60 radial bins on $|\omega| \in (0, 0.5]$. In each bin k, average to $\bar{P}_k$ and $\overline{|H|^2}_k = \mathrm{mean}|H|^2$.
3. Keep only **reliable** bins, where the signal clearly exceeds the noise: $\bar{P}_k > 2\hat{\sigma}^2$.
4. In each kept bin, remove the noise and undo the blur:

$$\hat{S}_k = \frac{\bar{P}_k - \hat{\sigma}^2}{\overline{|H|^2}_k}. \tag{24}$$

5. Fit

$$\log \hat{S}_k = \log A - \beta \log|\omega_k| \tag{25}$$

by least squares, which gives (A, β).

At least 3 reliable bins are required; otherwise the method stops with an error, and this never happened on our data. Restricting the fit to reliable bins is what keeps the de-blurring step in (4) stable: bins where noise dominates, and where division by $\overline{|H|^2}_k$ would explode, are never used.

C. The filter

$$\lambda(\omega) = \frac{\hat{\sigma}^2}{A\,|\omega|^{-\beta}} = \frac{\hat{\sigma}^2}{A}\,|\omega|^{\beta}, \tag{26}$$

$$\hat{X}(\omega) = \frac{H^*(\omega)\,Y(\omega)}{|H(\omega)|^2 + \lambda(\omega)}, \tag{27}$$

with $|\omega|$ at DC replaced by the smallest non-zero frequency, so the mean brightness passes unchanged. Compared with Tikhonov–Laplacian, where λ grows as $|\omega|^4$ with one scale chosen by GCV:

- the exponent β is fitted per image (2.67 on our development image);
- the scale $\hat{\sigma}^2/A$ comes from two physical quantities, the noise level and the signal power, rather than from a model-selection criterion.

Algorithm 1 (spectrally adaptive Wiener). Inputs: y ($M \times N$), PSF h. Parameters fixed once on development data: $\varepsilon = 10^{-3}$, SNR threshold 2, 60 bins.

1. $H \leftarrow$ DFT of zero-padded, centred h; $Y \leftarrow$ DFT of y.
2. $\hat{\sigma}^2 \leftarrow$ mean of $|Y|^2/(MN)$ over $\{|H|^2 < \varepsilon\}$ (wavelet MAD fallback if fewer than 50 bins).
3. For each radial bin with $\bar{P}_k > 2\hat{\sigma}^2$: $\hat{S}_k \leftarrow (\bar{P}_k - \hat{\sigma}^2)/\overline{|H|^2}_k$.
4. $(A, \beta) \leftarrow$ least-squares fit of $\log \hat{S}_k$ against $\log|\omega_k|$.
5. $\lambda(\omega) \leftarrow \hat{\sigma}^2\,|\omega|^\beta/A$.
6. Return $\hat{x} \leftarrow \mathrm{IDFT}\{H^*Y/(|H|^2 + \lambda)\}$.

For real images with non-periodic borders, y is first mirror-padded by 32 pixels and the result cropped back (Section 9.2).

D. Cost

The method needs one forward and one inverse 2-D FFT of the image, plus one FFT of the padded PSF: $O(MN \log MN)$ operations. The radial binning and the fit are $O(MN)$. The measured runtime on a local CPU was 10 ms per 256×256 image, against 2 ms for constant Wiener, 10 ms for Tikhonov–GCV (61 λ evaluations), 2 ms for Richardson–Lucy and 38 ms for unsupervised Wiener (Section 7, Table 4).

E. Sanity check on development data

On a development crop (WideField_BPAE_B, field 1; $\sigma_h = 2.5$, $\sigma_n = 0.02$):

- the estimated noise level was $\hat{\sigma} = 0.0200$, matching the true value to four decimals;
- the fitted exponent was $\beta = 2.67$;
- the method scored 29.83 dB PSNR, against 29.61 dB for the tuned constant-NSR Wiener filter and 30.30 dB for the oracle.

V. Baselines and Implementation

All methods share one implementation of the regularized inverse where possible. They also share the same PSF, OTF construction and boundary handling, so differences between them come only from how they regularize. Table 3 lists the methods compared.

Two baselines have a free parameter. The constant NSR and the RL iteration count were tuned once, on three development crops (WideField_BPAE_B, fields 1–3), averaged over every PSF width and noise level. They were then frozen in ‘configs/tuned.yaml‘ before any test image was processed.

- The constant NSR was searched over 17 log-spaced values in $[10^{-4}, 1]$; the best was 0.0316.
- The RL iteration count was searched over $\{1, 2, 3, 4, 5, 7, 10, 20, 30, 50\}$; the best was 1 (Section 5.1).

Implementation details.

- Images are float64 arrays in $[0, 1]$.
- The PSF is odd-sized ($8\sigma_h + 1$) and normalized to unit sum. The OTF is computed by ‘psf2otf‘ (zero-pad and circular shift).
- Outputs are clipped to $[0, 1]$ only for computing metrics; unclipped outputs are used for noise analysis.
- Random noise uses ‘numpy.random.default_rng(seed)‘ with seeds 0–4.
- All code is in ‘lib/‘ (‘forward.py‘, ‘data.py‘, ‘metrics.py‘, ‘baselines.py‘, ‘proposed.py‘).

Code verification:

- The inverse filter recovers a noiseless blurred image to a maximum error of 2.8×10^{-8}, which checks the OTF centring.
- On a test image, every method improves on the observation.
- The oracle Wiener filter has the highest PSNR of all methods at every setting (Table 9).

The implementation is shared across methods for PSF construction, OTF centring, boundary handling, and the regularized-inverse application wherever possible. The project uses float64 images normalized to $[0, 1]$, NumPy FFTs, and scikit-image implementations for Richardson–Lucy and unsupervised Wiener. Richardson–Lucy is used as provided by scikit-image (Section 5.1).

A. Richardson–Lucy tuning

The RL iteration count was tuned on the development images only (WideField_BPAE_B, FOVs 1–3, all PSF widths and noise levels) over the grid $\{1, 2, 3, 4, 5, 7, 10, 20, 30, 50\}$. The `skimage` implementation gave a mean development PSNR that fell with every extra iteration (30.32 dB at one iteration, 21.90 dB at 50), so the best setting sat at the grid minimum of one iteration. Since an iteration count cannot go below one, widening the grid would not change this. The `skimage` routine assumes zero-padded borders, whereas the benchmark blur is periodic like the model behind every FFT-based method here, which is a likely contributor to this behaviour. The final settings, frozen in `tuned.yaml` before any test image was processed, are NSR 0.0316 and RL with one iteration.

Starting from a flat image, the first RL iteration returns, away from the borders, the observed image blurred once more by the PSF, so at this early-stopped setting RL acts mainly as a mild smoother rather than a deconvolver. On the step-edge phantom ($\sigma_h = 2.5$ px) its edge FWHM is 8.36 px, wider than the 5.84 px of the unrestored blurred edge measured with the same noise levels and seeds; its MTF50 is 0.055 against 0.078, and its noise amplification is 0.45 against 1.00 (Section 7.5, Table 13). Its PSNR advantage over the unrestored image therefore comes mainly from noise suppression rather than recovered resolution, and the RL results in this report should be read in that light. We report the baseline as tuned and do not replace it with a stronger implementation, for example a mirror-padded one initialised from the observed image.

Table 4 summarizes the aggregate PSNR, SSIM, log-spectral distance and runtime of every method across the Gaussian benchmark.

VI. Experimental Setup

A. Datasets

Fluorescence Microscopy Denoising (FMD) [Zhang et al. 2019]. Real images from commercial confocal, two-photon and wide-field microscopes. For each field of view (FOV), 50 raw frames are recorded at the same position, and their average

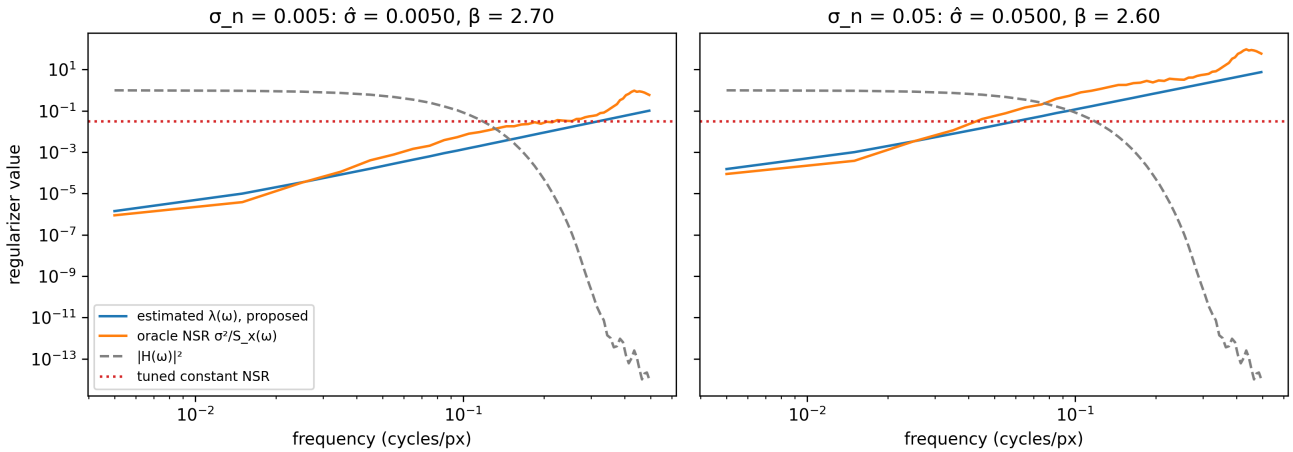


Fig. 2. Estimated frequency-dependent regularizer compared with the oracle noise-to-signal ratio, the OTF magnitude squared, and the tuned constant NSR at low and high Gaussian noise.

Table 3: Methods compared with the proposed filter: implementation, how the regularization is chosen, and reference.

Method	Implementation	Regularization chosen by	Reference
Inverse filter	$\hat{X} = H^*Y/(\lvert H\rvert^2 + 10^{-30})$	none	—
Constant-NSR Wiener	$\hat{X} = H^*Y/(\lvert H\rvert^2 + \mathrm{NSR})$	NSR tuned on development images (Section 6.2)	Wiener; Gonzalez & Woods
Tikhonov–GCV	Laplacian penalty, Section 3.2	GCV per image, 61 values of λ in $[10^{-6}, 1]$	Golub et al. 1979; Hansen et al. 2006
Richardson–Lucy (RL)	‘skimage.restoration.richardson_lucy‘ as provided (no extra padding), input clipped to $\geq 10^{-6}$	iteration count (1) tuned on development images	Richardson 1972; Lucy 1974
Unsupervised Wiener	‘skimage.restoration. unsupervised_wiener‘ (Laplacian prior, Gibbs sampling of noise and prior precisions), seeded	estimated per image by the sampler	Orieux et al. 2010
Oracle Wiener	$\lambda = \sigma_n^2/S_x(\omega)$, with S_x the periodogram of the true x	ground truth	upper bound only
Proposed	Algorithm 1	estimated per image	Section 4

Table 4: Aggregate PSNR, SSIM, log-spectral distance (LSD) and runtime of every method across the Gaussian benchmark (mean ± standard deviation over all test images, PSF widths, noise levels and seeds; runtime in milliseconds per image on a local CPU). RL is Richardson–Lucy with one iteration.

Method	PSNR (dB)	SSIM	LSD (dB)	Runtime (ms)
Oracle Wiener	30.767 ± 2.733	0.810 ± 0.070	26.390 ± 3.957	2.455 ± 0.300
Proposed	30.507 ± 2.692	0.785 ± 0.083	24.925 ± 3.994	9.552 ± 1.102
Tikhonov–GCV	30.520 ± 2.703	0.809 ± 0.069	26.778 ± 4.203	9.989 ± 1.119
Const.-NSR Wiener	29.528 ± 2.783	0.771 ± 0.148	27.979 ± 11.035	1.759 ± 0.193
RL	27.016 ± 2.828	0.801 ± 0.081	48.884 ± 8.891	1.758 ± 0.262
Unsup. Wiener	27.109 ± 4.113	0.464 ± 0.211	8.575 ± 0.839	38.391 ± 5.550
Inverse filter	3.372 ± 0.084	0.001 ± 0.000	22.961 ± 2.013	1.799 ± 0.209

(“avg50”) is provided as a near-noise-free reference. We use three subsets, each with 20 FOVs of 512×512 8-bit images (Table 5):

Licence: the FMD data release is CC BY-SA 4.0 (per the course sheet); the accompanying code is MIT-licensed. Source: github.com/yinhaoz/denoising-fluorescence. Downloaded on 29 Oct 2026 with gdown, using the Google-Drive file IDs in the authors’ download_dataset.sh. Only the raw/ and gt/ folders were extracted.

Kermany retinal OCT [Kermany et al. 2018]. Retinal OCT B-scans labelled CNV / DME / DRUSEN / NORMAL. We drew a seeded random sample (seed 0) of 20 files from the NORMAL and DRUSEN test folders of the Kaggle mirror ‘paultimothymooney/kermany2018‘.

- 7 of the 20 files are macOS metadata files (‘._*.jpeg‘, 212 bytes) packed in the archive, not images. They are excluded, which leaves 13 B-scans (4 DRUSEN, 9 NORMAL; e.g. 496×512 pixels).

B. Data partitions (fixed before any tuning)

Table 6 shows how the images are divided between development and test use.

The test FOV is a different microscope and sample type from the development subset, so the test also measures generalization across modalities. No parameter was changed

Table 5: Fluorescence-microscopy (FMD) subsets used in this study and the role of each.

Subset	Microscope	Sample	Role
WideField_BPAE_B	wide-field	fixed BPAE cells, nuclei channel (DAPI)	development
Confocal_BPAE_B	confocal	fixed BPAE cells, nuclei channel (DAPI)	test
TwoPhoton_MICE	two-photon	fixed mouse brain tissue (DAPI-stained, cleared)	test

Table 6: Division of the images into development, synthetic-test, real-test and OCT sets, fixed before any tuning.

Partition	Images	Used for
Development	WideField_BPAE_B, FOVs 1–18 (FOVs 1–3 used)	NSR and RL tuning; ε, SNR threshold and bin count of the proposed method; all debugging
Synthetic test	Confocal_BPAE_B and TwoPhoton_MICE, FOV 19 avg50, each cut into four non-overlapping 256×256 quadrants (8 crops)	every number in Sections 7–9
Real test	the same two FOVs: raw frame 0 versus avg50, centre 256×256 crop	Section 10.1
OCT	13 B-scans	Section 10.2 (qualitative)

after looking at test results. The split is recorded in configs/split.yaml.

Caveat on independence. The 8 synthetic test crops come from only 2 FOVs, and quadrants of one FOV share content and imaging conditions. The 480 paired comparisons per method (8 crops $\times$ 3 PSFs $\times$ 4 noise levels $\times$ 5 seeds) are therefore not independent, and the Wilcoxon p-values in Section 7 overstate the evidence. The per-setting means and standard deviations are the primary result.

C. Pre-processing

- Images are converted to float64 (RGB averaged to gray if present).
- Each image is scaled to $[0, 1]$ by the 0.5th and 99.5th percentiles of its clean reference, then clipped ('data.norm_pair'). The raw frame of the same FOV is scaled with the same two numbers, so the reference and the noisy frame stay on one intensity scale.
- After normalization, the real noise in FMD test field 19 (std of raw $-$ avg50, centre crop) is about 0.054 (confocal) and 0.094 (two-photon). These are higher than our largest synthetic $\sigma_n = 0.05$.
- OCT B-scans are transformed to log intensity, $z = \log(1+I)/\log(256)$, and are not cropped.

D. Degradation grid

Table 7 lists the degradations applied to the test images.

All parameters are in configs/params.yaml; the tuned values are in configs/tuned.yaml.

E. Metrics

Table 8 defines the metrics used in the experiments.

Statistics. Results are reported as mean ± standard deviation over crops and seeds. Paired comparisons use the two-sided Wilcoxon signed-rank test on per-trial PSNR (see the caveat in 6.2).

The synthetic test set comprises eight crops from two held-out fields of view. For Gaussian noise this gives $8 \times 3 \times 4 \times 5 = 480$ trials per method; for Poisson–Gaussian noise it gives $8 \times 3 \times 3 \times 5 = 360$ trials per method. These paired trials are not independent at the crop level because quadrants from a common FOV share content and acquisition conditions. Consequently, per-setting means and standard deviations are emphasized over extremely small paired-test p-values.

VII. Synthetic Experimental Results

A. Additive Gaussian noise: PSNR

Table 9 reports the updated PSNR values (mean ± standard deviation across the eight crops and five seeds). The proposed method remains close to the oracle Wiener upper bound across all PSF/noise settings. It is essentially tied with Tikhonov–GCV in PSNR; the paired median difference is only +0.004 dB, so no meaningful superiority over Tikhonov–GCV is claimed.

B. Structural similarity

Table 10 reports the corresponding SSIM values.

SSIM does not rank the methods identically to PSNR. Smoother outputs can receive higher SSIM in some low-noise settings even when their pixelwise fidelity is not the highest. This is clearest for RL, which has the highest SSIM of all methods in three settings (PSF 1.5 px at $\sigma_n = 0.01$ and 0.02, and PSF 2.5 px at $\sigma_n = 0.02$), despite the widest edges (Section 7.5); its aggregate SSIM (0.801) is nevertheless below the oracle (0.810) and Tikhonov–GCV (0.809). In particular, the unsupervised Wiener baseline degrades substantially at wider PSFs, consistent with its strong noise amplification.

C. Paired comparisons

Table 11 compares the proposed method with each baseline on paired trials.

For Gaussian noise, the median proposed-method gains are +0.716 dB over constant Wiener, +3.499 dB over Richardson–Lucy, and +3.110 dB over unsupervised Wiener; the proposed method beats RL in all paired trials. Against Tikhonov–GCV the median difference is +0.004 dB, with the proposed method

Table 7: Degradation grid: PSF widths, noise models and levels, noise seeds, boundary handling and PSF-mismatch settings.

Factor	Values
PSF	Gaussian, $\sigma_h \in \{1.5, 2.5, 4.0\}$ px
Gaussian noise	$\sigma_n \in \{0.005, 0.01, 0.02, 0.05\}$
Poisson–Gaussian noise	peak photons $p \in \{30, 100, 300\}$, read noise $\sigma_r = 0.01$
Noise seeds	0, 1, 2, 3, 4
Boundary	periodic (main), reflect (Section 9.2)
PSF mismatch	assumed $\sigma_h = 2.5(1+\delta)$, $\delta \in \{-30, \dots, +30\}$ %, true 2.5 px, $\sigma_n = 0.01$

Table 8: Evaluation metrics, their definitions, and the sections in which each is used.

Metric	Definition	Used in
PSNR (dB)	$10 \log_{10}(1/\mathrm{MSE})$ on outputs clipped to $[0, 1]$	7, 8, 9
SSIM	'skimage.metrics.structural_similarity', data range 1	7, 8
Log-spectral distance (dB)	RMS over ω of $10 \log_{10}(\lvert X\rvert^2/\lvert\hat{X}\rvert^2)$	7
Edge FWHM (px)	full width at half maximum of the line-spread function (derivative of the row-averaged edge profile of a step-edge phantom, $0.2 \to 0.8$)	7.5
MTF50 (cycles/px)	frequency where the Fourier magnitude of the ±16 px LSF window drops to 0.5	7.5
Overshoot (%)	peak of the edge profile above the bright level, as % of the step height (ringing)	7.5
Noise amplification (synthetic)	std in the flat half of the restored edge phantom ÷ σ_n	7.5
Noise gain (real)	std$(f(\text{raw}) - f(\text{avg50}))$ / std(raw − avg50), with the same filter f applied to both (Section 10.1)	10
Chain MTF50 (real)	MTF50 of $\lvert H\rvert^2/(\lvert H\rvert^2 + \lambda)$, the transfer of blur followed by restoration	10
Runtime (ms)	wall-clock per 256×256 image, local CPU	7

Table 9: Gaussian-noise PSNR (dB) of each method for every PSF width and noise level (mean ± standard deviation over crops and seeds). RL is Richardson–Lucy with one iteration.

σ_h (px)	σ_n	Const. Wiener	Tikhonov–GCV	RL	Unsup. Wiener	Proposed	Oracle
1.5	0.005	33.75 ± 1.69	34.53 ± 1.51	29.62 ± 2.22	33.91 ± 1.33	34.47 ± 1.53	34.79 ± 1.54
1.5	0.01	33.05 ± 1.46	33.51 ± 1.52	29.60 ± 2.20	32.62 ± 1.22	33.44 ± 1.55	33.79 ± 1.55
1.5	0.02	31.30 ± 1.06	32.32 ± 1.49	29.52 ± 2.14	31.06 ± 1.07	32.26 ± 1.54	32.63 ± 1.54
1.5	0.05	26.77 ± 0.48	30.43 ± 1.50	28.79 ± 1.65	28.45 ± 0.80	30.41 ± 1.51	30.80 ± 1.51
2.5	0.005	30.52 ± 1.98	31.85 ± 1.69	26.98 ± 2.25	29.30 ± 0.96	31.82 ± 1.72	32.04 ± 1.72
2.5	0.01	30.38 ± 1.91	31.08 ± 1.70	26.97 ± 2.24	28.12 ± 0.84	31.06 ± 1.71	31.30 ± 1.71
2.5	0.02	29.92 ± 1.73	30.18 ± 1.71	26.94 ± 2.21	26.75 ± 0.71	30.18 ± 1.72	30.44 ± 1.71
2.5	0.05	28.02 ± 1.19	28.79 ± 1.71	26.55 ± 1.92	24.56 ± 0.54	28.83 ± 1.71	29.11 ± 1.70
4.0	0.005	27.89 ± 2.09	29.33 ± 1.89	24.88 ± 2.22	24.35 ± 0.59	29.34 ± 1.90	29.49 ± 1.89
4.0	0.01	27.86 ± 2.07	28.77 ± 1.88	24.88 ± 2.22	23.35 ± 0.52	28.78 ± 1.89	28.95 ± 1.89
4.0	0.02	27.74 ± 2.01	28.17 ± 1.87	24.86 ± 2.20	22.25 ± 0.45	28.19 ± 1.88	28.36 ± 1.88
4.0	0.05	27.13 ± 1.75	27.29 ± 1.83	24.61 ± 2.02	20.59 ± 0.37	27.31 ± 1.86	27.50 ± 1.86

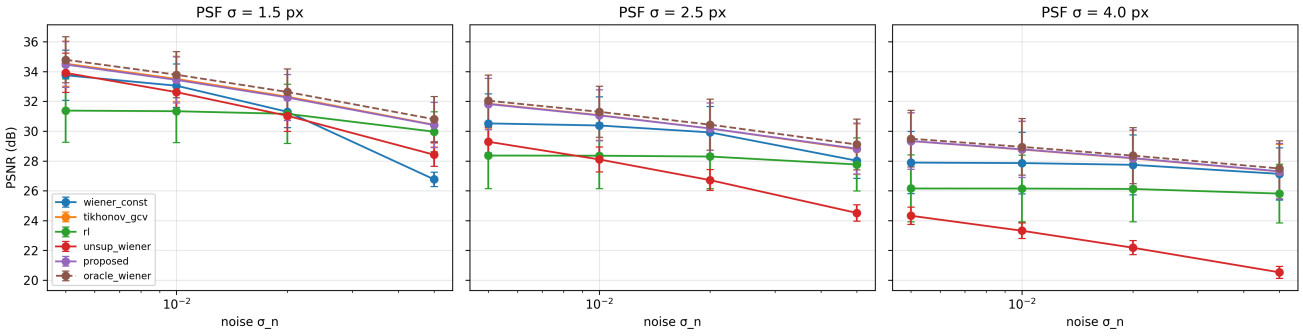


Fig. 3. Gaussian-noise PSNR versus noise standard deviation for the three Gaussian PSF widths. Error bars show variation across test crops and seeds.

winning 54.4% of paired trials. Given the correlated crop structure, the near-zero effect size is the important result; the small p-value should not be interpreted as a practically meaningful improvement.

D. Poisson–Gaussian noise

Under signal-dependent Poisson–Gaussian noise, the proposed method retains gains over the tuned constant Wiener filter (+0.471 dB median), Richardson–Lucy (+2.637 dB), and unsupervised Wiener (+4.118 dB). The proposed method wins all paired trials against RL, and RL does not score higher in any of the nine PSF/photon settings. Tikhonov–GCV has

Table 10: Gaussian-noise SSIM of each method for every PSF width and noise level (mean ± standard deviation).

σ_h (px)	σ_n	Const. Wiener	Tikhonov–GCV	RL	Unsup. Wiener	Proposed	Oracle
1.5	0.005	0.916 ± 0.032	0.897 ± 0.036	0.890 ± 0.044	0.838 ± 0.030	0.882 ± 0.044	0.899 ± 0.035
1.5	0.01	0.861 ± 0.028	0.872 ± 0.040	0.882 ± 0.039	0.773 ± 0.026	0.852 ± 0.050	0.876 ± 0.038
1.5	0.02	0.719 ± 0.020	0.836 ± 0.044	0.852 ± 0.020	0.679 ± 0.020	0.812 ± 0.057	0.841 ± 0.041
1.5	0.05	0.406 ± 0.030	0.765 ± 0.041	0.734 ± 0.060	0.516 ± 0.018	0.738 ± 0.063	0.772 ± 0.045
2.5	0.005	0.895 ± 0.042	0.860 ± 0.048	0.847 ± 0.053	0.574 ± 0.018	0.841 ± 0.059	0.861 ± 0.046
2.5	0.01	0.872 ± 0.038	0.835 ± 0.050	0.840 ± 0.048	0.498 ± 0.020	0.814 ± 0.062	0.835 ± 0.049
2.5	0.02	0.802 ± 0.032	0.803 ± 0.050	0.813 ± 0.030	0.416 ± 0.024	0.779 ± 0.064	0.802 ± 0.051
2.5	0.05	0.584 ± 0.017	0.745 ± 0.042	0.709 ± 0.060	0.310 ± 0.026	0.719 ± 0.065	0.746 ± 0.051
4.0	0.005	0.857 ± 0.051	0.815 ± 0.052	0.803 ± 0.060	0.300 ± 0.025	0.795 ± 0.064	0.809 ± 0.054
4.0	0.01	0.846 ± 0.046	0.797 ± 0.053	0.797 ± 0.055	0.261 ± 0.025	0.770 ± 0.066	0.790 ± 0.054
4.0	0.02	0.812 ± 0.038	0.773 ± 0.052	0.772 ± 0.041	0.226 ± 0.023	0.738 ± 0.067	0.766 ± 0.054
4.0	0.05	0.684 ± 0.028	0.711 ± 0.037	0.673 ± 0.062	0.184 ± 0.019	0.683 ± 0.068	0.722 ± 0.053

Table 11: Paired Wilcoxon signed-rank comparisons of the proposed method against each baseline; gain is proposed minus baseline PSNR.

Noise model	Baseline	n	Median gain (dB)	Mean gain (dB)	Proposed wins	p-value
Gaussian	Const.-NSR Wiener	480	0.716	0.979	1.000	2.3×10^{-80}
Gaussian	Tikhonov–GCV	480	0.004	−0.013	0.544	0.047
Gaussian	RL	480	3.499	3.491	1.000	2.3×10^{-80}
Gaussian	Unsup. Wiener	480	3.110	3.398	1.000	2.3×10^{-80}
Poisson–Gaussian	Const.-NSR Wiener	360	0.471	1.091	0.983	1.3×10^{-60}
Poisson–Gaussian	Tikhonov–GCV	360	−0.015	−0.046	0.319	5.9×10^{-21}
Poisson–Gaussian	RL	360	2.637	2.479	1.000	9.4×10^{-61}
Poisson–Gaussian	Unsup. Wiener	360	4.118	4.257	1.000	9.4×10^{-61}

Table 12: Poisson–Gaussian PSNR (dB) for every PSF width and photon level (mean ± standard deviation); peak is the photon level, and larger means less noise. RL is Richardson–Lucy with one iteration.

σ_h (px)	Peak photons	Const. Wiener	Tikhonov–GCV	RL	Unsup. Wiener	Proposed	Oracle
1.5	30	26.01 ± 1.27	30.37 ± 1.66	29.27 ± 2.10	27.90 ± 1.05	30.28 ± 1.69	30.76 ± 1.69
1.5	100	29.42 ± 1.24	31.58 ± 1.61	29.50 ± 2.16	29.76 ± 1.12	31.49 ± 1.66	31.93 ± 1.65
1.5	300	31.43 ± 1.30	32.45 ± 1.59	29.57 ± 2.19	31.04 ± 1.17	32.36 ± 1.63	32.76 ± 1.63
2.5	30	27.61 ± 1.41	28.75 ± 1.80	26.89 ± 2.21	23.83 ± 0.72	28.73 ± 1.81	29.06 ± 1.81
2.5	100	29.32 ± 1.70	29.65 ± 1.80	26.95 ± 2.23	25.37 ± 0.82	29.63 ± 1.82	29.92 ± 1.81
2.5	300	30.00 ± 1.82	30.29 ± 1.77	26.97 ± 2.24	26.51 ± 0.81	30.27 ± 1.78	30.54 ± 1.78
4.0	30	27.08 ± 1.88	27.27 ± 1.93	24.85 ± 2.22	20.09 ± 0.57	27.23 ± 1.97	27.49 ± 1.96
4.0	100	27.60 ± 2.01	27.84 ± 1.92	24.87 ± 2.22	21.20 ± 0.60	27.82 ± 1.97	28.03 ± 1.96
4.0	300	27.77 ± 2.05	28.25 ± 1.92	24.88 ± 2.22	22.03 ± 0.56	28.25 ± 1.93	28.45 ± 1.93

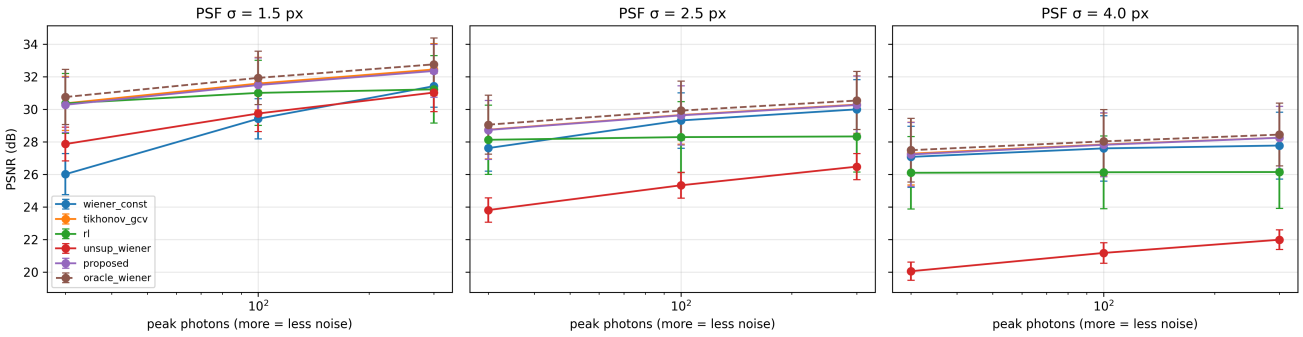


Fig. 4. PSNR under Poisson–Gaussian noise for photon levels 30, 100, and 300 and the three PSF widths.

a small median advantage of 0.015 dB. These results are consistent with the limitation that the proposed filter estimates one global noise power and therefore cannot represent local brightness-dependent variance exactly. Table 12 reports PSNR under Poisson–Gaussian noise.

E. Resolution, ringing, and noise amplification

Table 13 and Table 14 report resolution, ringing and noise amplification measured on a step-edge phantom.

The updated edge metrics show that the proposed method and Tikhonov–GCV have very similar aggregate FWHM (4.813 and 4.804 px) and MTF50 (both 0.127 cycles/pixel),

Table 13: Aggregate edge-phantom metrics (mean ± standard deviation): edge FWHM in pixels, MTF50 in cycles/pixel, overshoot in percent, and noise amplification (output noise standard deviation divided by input σ_n).

Method	FWHM (px)	MTF50 (cycles/px)	Overshoot (%)	Noise amplification
Oracle Wiener	3.176 ± 0.294	0.190 ± 0.018	9.382 ± 1.532	0.636 ± 0.493
Proposed	4.813 ± 0.777	0.127 ± 0.021	7.374 ± 0.843	0.879 ± 0.536
Tikhonov–GCV	4.804 ± 0.797	0.127 ± 0.021	7.582 ± 0.711	0.888 ± 0.551
Const.-NSR Wiener	4.806 ± 0.057	0.121 ± 0.002	1.312 ± 0.371	0.620 ± 0.015
RL	8.359 ± 0.020	0.055 ± 0.000	0.103 ± 0.102	0.448 ± 0.306
Unsup. Wiener	3.617 ± 0.294	0.164 ± 0.014	7.738 ± 0.645	3.389 ± 1.814

Table 14: Edge-phantom metrics broken down by Gaussian noise level (means).

σ_n	Method	FWHM (px)	MTF50	Overshoot (%)	Noise amp.
0.005	Oracle Wiener	2.848	0.212	9.847	1.326
0.005	**Proposed**	3.951	0.152	8.278	1.684
0.005	RL	8.346	0.055	0.024	0.922
0.005	Tikhonov–GCV	3.918	0.153	8.333	1.726
0.005	Unsup. Wiener	3.291	0.179	8.470	6.065
0.005	Const.-NSR Wiener	4.817	0.121	1.472	0.620
0.01	Oracle Wiener	3.026	0.198	10.070	0.688
0.01	**Proposed**	4.391	0.138	7.842	0.964
0.01	RL	8.350	0.055	0.048	0.471
0.01	Tikhonov–GCV	4.393	0.138	7.989	0.949
0.01	Unsup. Wiener	3.471	0.170	7.599	3.759
0.01	Const.-NSR Wiener	4.814	0.121	1.417	0.620
0.02	Oracle Wiener	3.236	0.184	9.554	0.366
0.02	**Proposed**	4.930	0.121	7.161	0.571
0.02	RL	8.358	0.055	0.095	0.254
0.02	Tikhonov–GCV	4.897	0.122	7.351	0.583
0.02	Unsup. Wiener	3.678	0.161	7.633	2.385
0.02	Const.-NSR Wiener	4.807	0.122	1.309	0.620
0.05	Oracle Wiener	3.594	0.167	8.058	0.162
0.05	**Proposed**	5.978	0.097	6.214	0.297
0.05	RL	8.382	0.055	0.246	0.143
0.05	Tikhonov–GCV	6.007	0.097	6.656	0.295
0.05	Unsup. Wiener	4.028	0.145	7.249	1.347
0.05	Const.-NSR Wiener	4.788	0.122	1.051	0.620

with approximately 7.4–7.6% overshoot. The noise-dependent table shows the expected adaptive trade-off: at $\sigma_n = 0.005$, the proposed method achieves 3.951 px FWHM and 0.152 cycles/pixel MTF50 while amplifying noise by 1.684 times; at $\sigma_n = 0.05$, it uses a more conservative response (5.978 px, 0.097 cycles/pixel) and reduces noise to 0.297 times. The constant Wiener filter holds a nearly fixed compromise across noise levels. RL has the widest edges (8.36 px, against 5.84 px measured for the unrestored blurred edge with the same noise levels and seeds), an MTF50 of 0.055 (unrestored: 0.078), 0.1% overshoot and a noise amplification of 0.45, which is consistent with its role as a smoother at its tuned setting (Section 5.1).

F. Runtime

Table 4 reports 9.552 ± 1.102 ms per image for the proposed method, compared with 1.759 ± 0.193 ms for constant Wiener, 9.989 ± 1.119 ms for Tikhonov–GCV, 1.758 ± 0.262 ms for Richardson–Lucy, 38.391 ± 5.550 ms for unsupervised Wiener, and 2.455 ± 0.300 ms for the oracle Wiener filter. These are measured values from the aggregate CSV on a local CPU; runtime is environment-dependent.

VIII. Ablation Study

The proposed $\lambda(\omega)$ has two estimated ingredients: the noise level $\hat{\sigma}^2$ (Section 4.1) and the signal-spectrum model $A|\omega|^{-\beta}$ (Section 4.2). To see which one limits performance, we replace each with its true value, one at a time. To check whether the frequency shaping matters at all, we also collapse $\lambda(\omega)$ to a single constant. Table 15 defines the variants.

All variants run on the same 480 degraded test images as Section 7, using the same seeds. The "full" variant must therefore reproduce the proposed method's numbers exactly; and the largest absolute PSNR difference between the full variant is 7.1×10^{-15} dB, i.e. floating-point rounding.

A positive value means the true quantity does better, so that estimate is the bottleneck. A negative constant-λ value means the frequency shaping is worth that many dB.

Expected outcome, stated before the run so the result can confirm or refute it:

- Oracle $\sigma \approx 0$ dB. The noise estimate was exact to four decimals on development data ($\hat{\sigma} = 0.0200$ for $\sigma_n = 0.02$). Table 17 ('table9_estimator_accuracy.csv') reports $\hat{\sigma}/\sigma_n$ over all settings.
- Oracle $S_x \approx +0.15$ to $+0.39$ dB, close to the whole proposed-to-oracle gap in Table 4. That would locate the

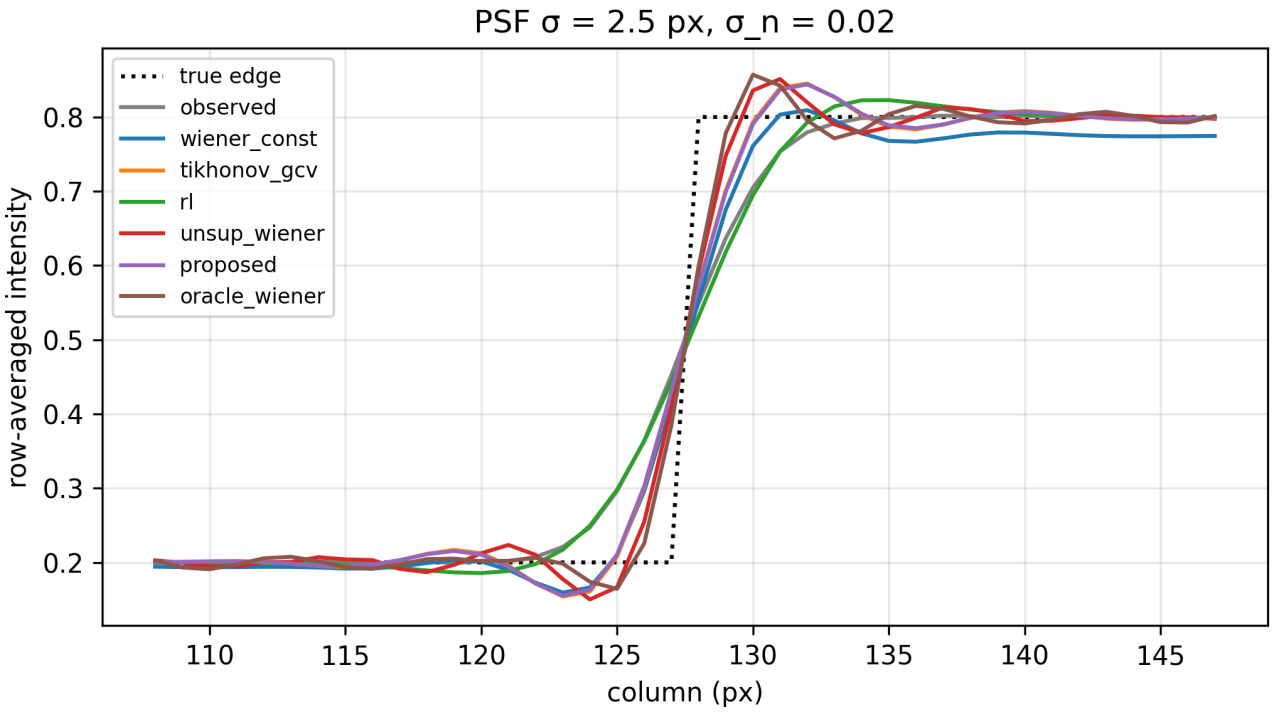


Fig. 5. Row-averaged edge profiles for the true step, observed blurred/noisy edge, and the restoration methods at PSF $\sigma_h = 2.5$ px and noise standard deviation $\sigma_n = 0.02$.

Table 15: Ablation variants: which ingredient of $\lambda(\omega)$ is replaced by its true value, or collapsed to a constant.

Variant	Noise level	Signal spectrum	λ
Full (proposed)	$\hat{\sigma}^2$ from the noise band	power-law fit	$\hat{\sigma}^2/(A\,\lvert\omega\rvert^{-\beta})$
Oracle σ	true σ_n^2	power-law fit (refitted with σ_n^2)	$\sigma_n^2/(A\,\lvert\omega\rvert^{-\beta})$
Oracle S_x	$\hat{\sigma}^2$	true periodogram of x	$\hat{\sigma}^2/S_x(\omega)$
Constant λ	$\hat{\sigma}^2$	power-law fit	geometric mean of the full $\lambda(\omega)$, one value for all ω

remaining gap in the isotropic power-law model, which cannot represent the anisotropic, oriented structure of real cell images.

- Constant $\lambda < 0$. If the shaping matters, collapsing $\lambda(\omega)$ should cost about as much as the gap between the proposed method and the constant-NSR Wiener filter.
- the mean and spread of $\hat{\sigma}/\sigma_n$ for each PSF and noise level;
- the range of fitted β.

Expected: $\hat{\sigma}/\sigma_n \approx 1$ when the noise-only band is large ($\sigma_h \geq 2.5$). There may be a slight over-estimate at $\sigma_h = 1.5$, $\sigma_n = 0.005$, where the band starts at 0.28 cycles/px and residual signal can leak in.

Table 16 reports the measured PSNR change for each variant.

Table 17 reports how accurately the noise level and the spectral exponent are estimated.

The oracle-noise ablation changes PSNR by approximately 0 dB at all PSF widths, indicating that the noise-band estimate is highly accurate on this controlled Gaussian benchmark. Replacing the fitted spectrum with the true signal periodogram improves PSNR by 0.36, 0.25, and 0.17 dB for PSF widths 1.5, 2.5, and 4.0 px, respectively. Collapsing $\lambda(\omega)$ to a constant reduces PSNR by 4.91, 3.98, and 3.10 dB, showing that frequency shaping is a major contributor. The fitted β decreases from roughly 2.57 to 2.11 across the tabulated conditions as the blur width and noise level increase; the table reports the condition-wise means and standard deviations.

IX. Robustness and Boundary Handling

A. PSF-width mismatch

The proposed method, Tikhonov–GCV and constant Wiener peak near the correctly specified PSF width, whereas RL changes little over the range, decreasing slowly from 27.61 dB at −30% to 26.34 dB at +30%, as expected of a mild smoother. The proposed method decreases from 31.05 dB at zero mismatch to 27.22 dB at +30% mismatch; Tikhonov–GCV decreases from 31.07 to 25.84 dB over the same positive mismatch range. The proposed method is therefore less degraded than Tikhonov–GCV for an over-wide assumed PSF; at +30% it is still above RL (26.34 dB) but falls below constant Wiener (29.95 dB). Constant Wiener is the most robust of the compared methods to PSF misspecification, illustrating a trade-off: a fixed, conservative filter can tolerate PSF error

Table 16: PSNR change relative to the full proposed method (dB), mean ± standard deviation, grouped by PSF width.

σ_h (px)	Oracle σ	Oracle S_x	Constant λ
1.5	0.00 ± 0.00	0.36 ± 0.07	−4.91 ± 4.66
2.5	0.00 ± 0.00	0.25 ± 0.05	−3.98 ± 4.00
4	0.00 ± 0.00	0.17 ± 0.03	−3.10 ± 3.29

Table 17: Estimator accuracy: estimated noise standard deviation divided by true σ_n, and fitted power-law exponent β (mean ± standard deviation).

σ_h (px)	σ_n	$\hat{\sigma}/\sigma_n$	β
1.5	0.005	0.999 ± 0.001	2.570 ± 0.231
1.5	0.01	0.998 ± 0.001	2.511 ± 0.256
1.5	0.02	0.998 ± 0.001	2.448 ± 0.286
1.5	0.05	0.998 ± 0.001	2.357 ± 0.338
2.5	0.005	1.000 ± 0.002	2.403 ± 0.314
2.5	0.01	0.999 ± 0.002	2.369 ± 0.338
2.5	0.02	0.999 ± 0.002	2.333 ± 0.354
2.5	0.05	0.999 ± 0.002	2.276 ± 0.392
4	0.005	1.001 ± 0.002	2.295 ± 0.386
4	0.01	0.999 ± 0.001	2.257 ± 0.399
4	0.02	0.999 ± 0.001	2.208 ± 0.416
4	0.05	0.998 ± 0.001	2.107 ± 0.452

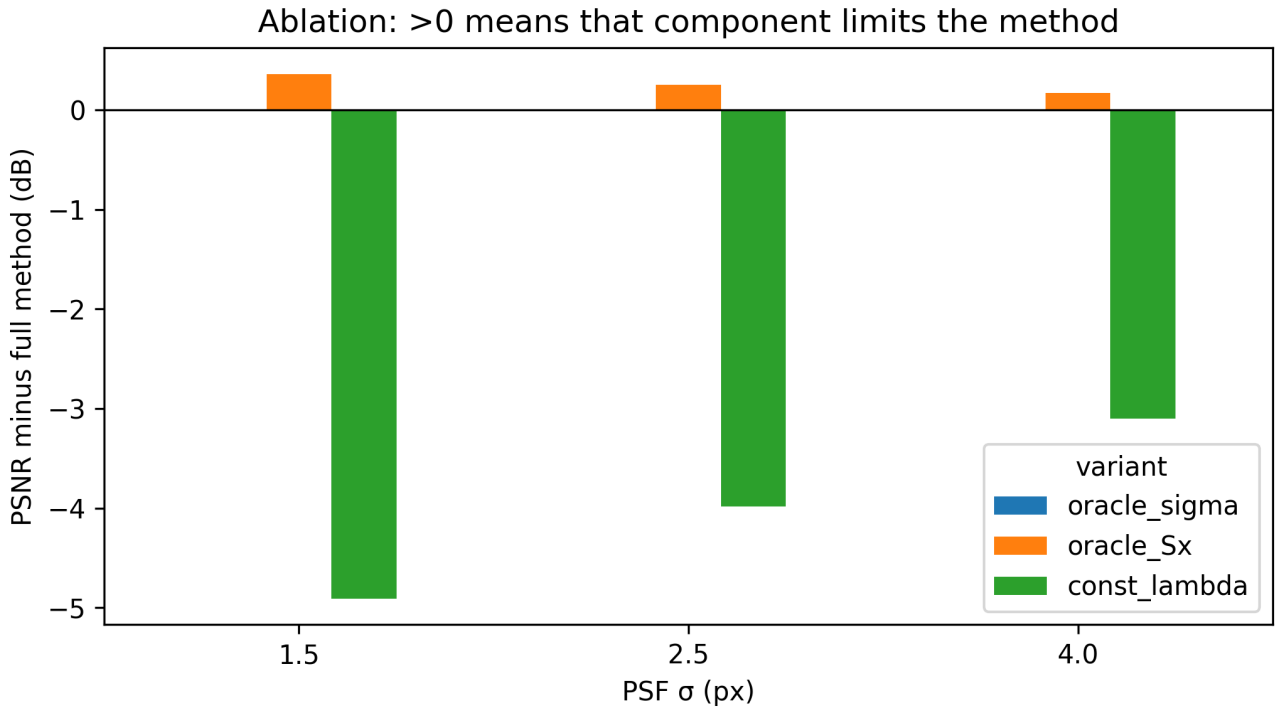


Fig. 6. Ablation of the estimated noise level, estimated signal spectrum, and frequency-dependent regularization.

Table 18: PSNR (dB) versus error in the assumed PSF width (mean ± standard deviation over 8 crops and 3 seeds, i.e. 24 trials per cell). RL is Richardson–Lucy with one iteration.

PSF width error	Proposed	RL	Tikhonov–GCV	Const. Wiener
−30 %	29.77 ± 1.94	27.61 ± 2.21	29.63 ± 1.94	29.34 ± 1.92
−20 %	30.25 ± 1.87	27.41 ± 2.23	30.14 ± 1.88	29.79 ± 1.92
−10 %	30.75 ± 1.78	27.19 ± 2.24	30.69 ± 1.79	30.16 ± 1.92
0 %	31.05 ± 1.73	26.97 ± 2.26	31.07 ± 1.71	30.38 ± 1.93
+10 %	30.74 ± 1.80	26.76 ± 2.27	30.73 ± 1.76	30.41 ± 1.94
+20 %	29.42 ± 2.08	26.55 ± 2.28	28.97 ± 2.12	30.26 ± 1.96
+30 %	27.22 ± 2.42	26.34 ± 2.29	25.84 ± 2.59	29.95 ± 1.98

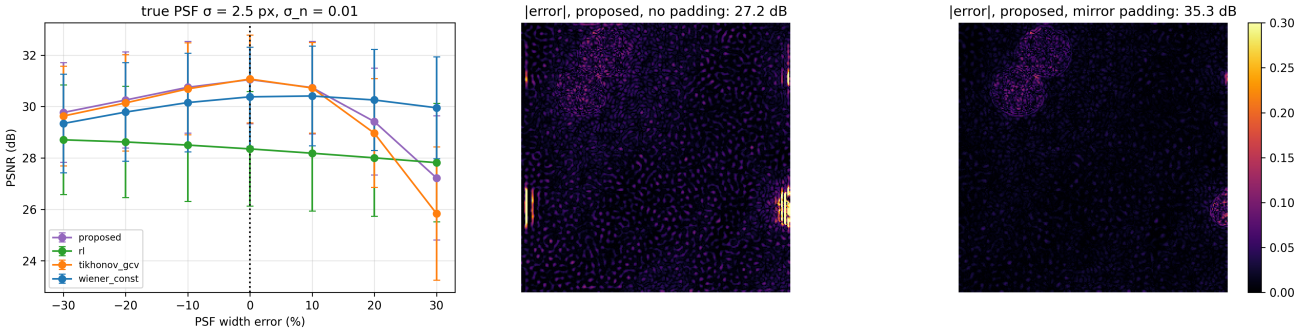


Fig. 7. PSF-width mismatch and boundary-handling robustness. The left panel shows PSNR as the assumed PSF width varies; the other panels illustrate boundary error with and without mirror padding.

better, while the proposed adaptive method targets a stronger frequency-dependent restoration when its model is appropriate. The method's noise-only band is also defined by the assumed OTF: an assumed PSF that is too wide can admit residual signal into the band and overestimate noise, making the filter more conservative. Each value is a mean over 8 crops and 3 seeds (24 trials); the standard deviations, 1.7–2.6 dB, are dominated by differences between crops, so the differences between methods should be read from the paired trends rather than from single cells. Table 18 reports PSNR as the assumed PSF width departs from the true width.

B. Boundary handling

For the proposed method, the aggregate PSNR rises from 25.95 ± 2.86 dB without padding to 28.58 ± 3.43 dB with mirror padding. A separate representative-crop illustration in the supplied figure reports 27.2 dB without padding and 35.3 dB with mirror padding. The aggregate metrics report the mean across all test crops/seeds, whereas the figure annotation calls out a specific single representative crop. These values refer to different scopes and are not contradictory. Mirror padding is therefore used for the real-image evaluations. Table 19 compares reconstruction with and without mirror padding.

X. Real-Image Evaluation

The real FMD and OCT experiments do not have an exact clean ground truth or a measured PSF. Accordingly, this section reports noise-gain measurements, chain MTF50, background-noise ratios, and qualitative image comparisons. It does not claim objectively verified reconstruction accuracy or that newly visible structures are physically true.

A. Fluorescence microscopy (FMD)

The proposed method's measured noise gain ranges from 0.603 to 1.469 across the two FMD subsets and the three assumed PSF widths. Tikhonov–GCV reaches $4.29\times$ on the confocal field at assumed $\sigma_h = 2.0$ px, while the proposed method is $1.469\times$ in that condition. The proposed method's chain MTF50 ranges from 0.162 to 0.208 cycles/pixel. RL, run on the same mirror-padded images as the other methods, has the lowest noise gain of all methods in every condition (0.27–0.48), consistent with its role as a smoother (Section 5.1). These measurements support a more moderate noise/resolution trade-off under the real acquisition noise, but they do not establish absolute reconstruction fidelity. Table 20 reports the real-noise gain and chain MTF50 for each held-out field and assumed PSF width.

B. Retinal OCT

The proposed method has mean OCT background-noise ratio 1.02 (standard deviation 0.205; maximum 1.252) and mean lateral MTF50 0.179 cycles/pixel. Tikhonov–GCV has mean background-noise ratio 3.897, with a large standard deviation of 6.957 and maximum 21.97; its mean lateral MTF50 is 0.204 cycles/pixel. Constant Wiener has mean noise ratio 1.453 and mean lateral MTF50 0.212 cycles/pixel. The OCT evaluation is qualitative: log-speckle is non-Gaussian and spatially correlated, the lateral PSF is assumed, and the vitreous band itself contains speckle rather than pure detector noise. Table 21 reports the OCT background-noise ratio and lateral MTF50.

XI. Discussion and Limitations

A. What the results show

1. Frequency-dependent regularization estimated from the data is close to optimal on matched synthetic data. The proposed $\lambda(\omega)$ comes within 0.15–0.39 dB of the oracle Wiener filter without any tuning. One constant NSR, by contrast, costs up to 3.6 dB at the noise levels it was not tuned for (Table 9).

2. On synthetic data the gain over a well-chosen Tikhonov regularizer is nil. GCV with a Laplacian penalty, which implies the spectral model $S_x \propto |\omega|^{-4}$, performs as well as our fitted $|\omega|^{-\beta}$ model when the noise really is white and the PSF is exact. We do not claim a PSNR improvement over Tikhonov–GCV.

B. Failure modes and why they occur (DSP view)

- Ringing. The methods that sharpen most (proposed, Tikhonov–GCV, unsupervised Wiener and the oracle) overshoot by about 7–9 % at a step edge (Section 7.5, Table 13); the conservative constant Wiener filter (1.3 %) and the smoothing RL (0.1 %) overshoot less. The restoration transfer function $T(\omega) = |H|^2/(|H|^2 + \lambda)$ falls

Table 19: Aggregate PSNR (dB) with and without mirror padding at the image boundary (mean ± standard deviation). Values are over 8 crops (one seed each).

Method	No padding	Mirror padding
Proposed	25.95 ± 2.86	28.58 ± 3.43
Tikhonov–GCV	14.41 ± 8.75	17.48 ± 9.00
Const.-NSR Wiener	30.24 ± 1.98	30.91 ± 2.39

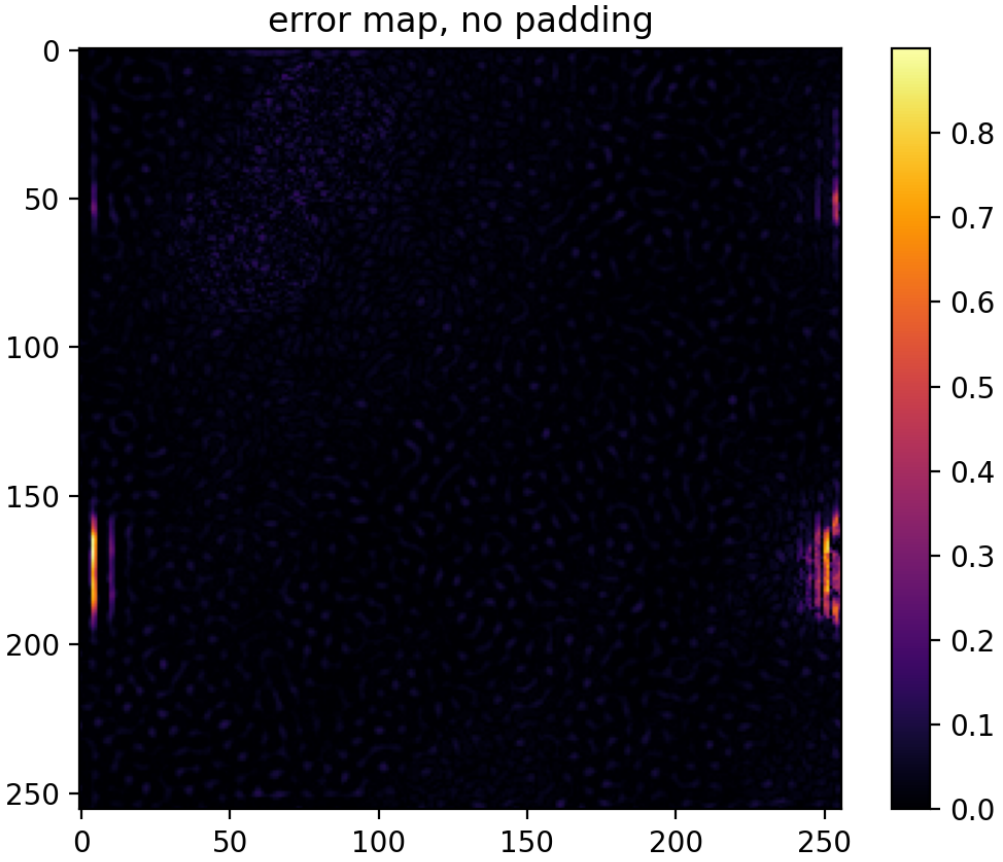


Fig. 8. Absolute reconstruction error without padding. The error is concentrated near the image boundaries, consistent with the periodic-boundary assumption of the FFT implementation.

Table 20: Real-noise gain and chain MTF50 on the held-out FMD fields for each assumed Gaussian PSF width (MTF50 in cycles/pixel; noise gain is dimensionless).

Subset	σ_h (px)	Chain MTF50 (cycles/px)				Noise gain				
		Observed	Proposed	Tik.–GCV	Const.	Observed	Proposed	RL	Tik.–GCV	Const.
Confocal_BPAE_B	1	0.188	0.208	0.218	0.298	1.000	0.681	0.478	0.726	1.510
Confocal_BPAE_B	1.5	0.128	0.188	0.208	0.198	1.000	0.938	0.416	1.235	1.135
Confocal_BPAE_B	2	0.098	0.172	0.212	0.152	1.000	1.469	0.382	4.290	0.955
TwoPhoton_MICE	1	0.188	0.202	0.198	0.298	1.000	0.603	0.373	0.582	1.557
TwoPhoton_MICE	1.5	0.128	0.178	0.188	0.198	1.000	0.756	0.308	0.866	1.115
TwoPhoton_MICE	2	0.098	0.162	0.182	0.152	1.000	1.110	0.274	1.701	0.904

from ≈ 1 to ≈ 0 over a narrow band of frequencies. A sharp cut-off in frequency produces oscillations in space (the Gibbs phenomenon). A smoother roll-off would trade ringing for resolution.

- Coloured residual noise. Restored noise is not white: it is white noise shaped by $|G(\omega)|^2$, which peaks just below the cut-off. This produces the "worm-like" texture in Fig. 10. Noise standard deviation alone understates how visible it is.
- Boundaries. Without padding, border discontinuities cost

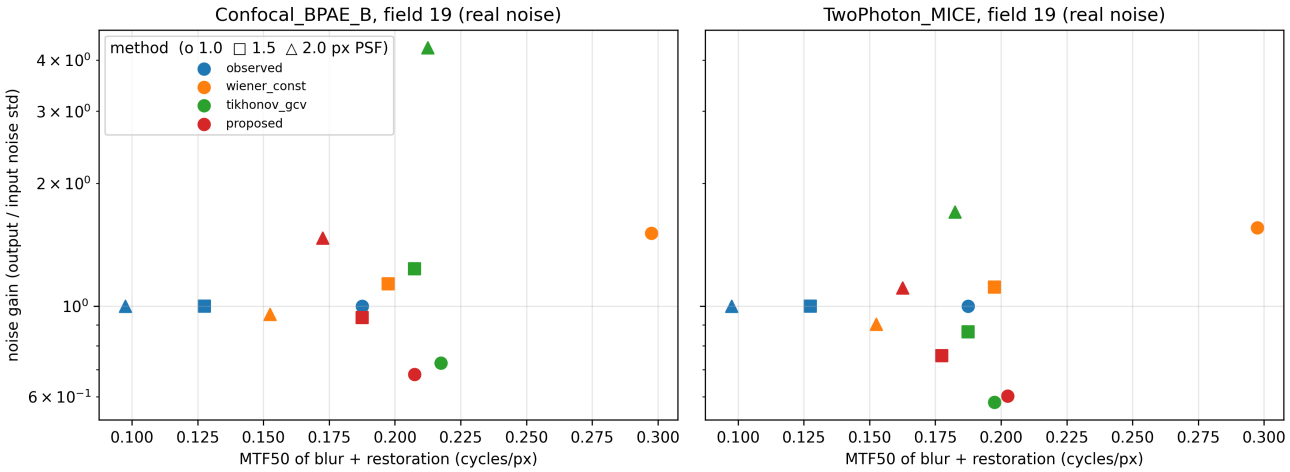


Fig. 9. Trade-off between real-noise gain and chain MTF50 for the Confocal BPAE and Two-Photon MICE FMD fields.

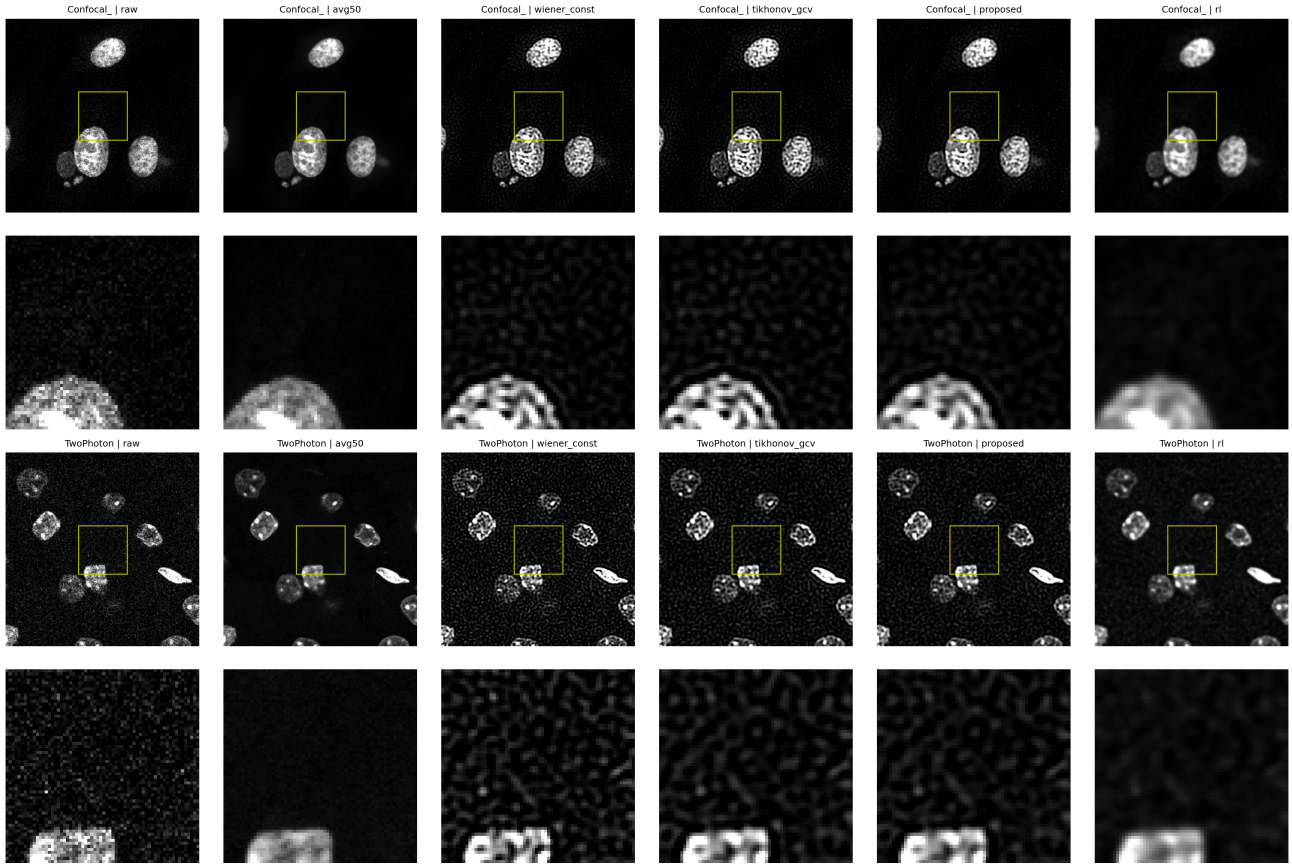


Fig. 10. FMD raw frame, 50-frame average, and deconvolution outputs, with corresponding zoomed regions. These panels are qualitative; the average is a near-noise-free reference, not an exact noiseless ground truth.

Table 21: OCT background-noise ratio and lateral MTF50 across the 13 usable B-scans (mean, standard deviation and maximum, where supplied).

Method	Background-noise ratio			Lateral MTF50 (cycles/px)		
	Mean	Std	Max	Mean	Std	Max
Proposed	1.020	0.205	1.252	0.179	0.014	0.192
Tikhonov–GCV	3.897	6.957	21.970	0.204	0.064	0.348
Const.-NSR Wiener	1.453	0.079	1.543	0.212	0.000	0.212

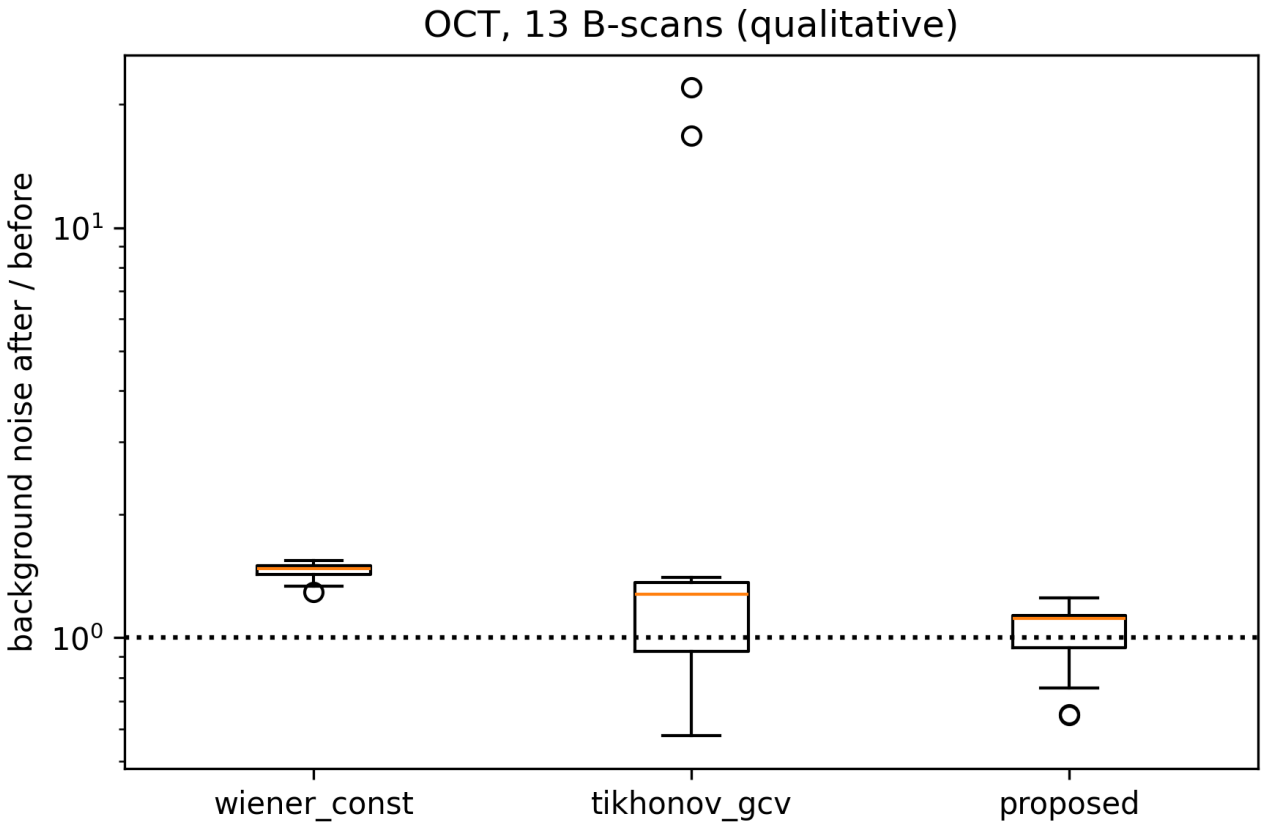


Fig. 11. OCT background-noise ratio after versus before restoration across the evaluated B-scans.

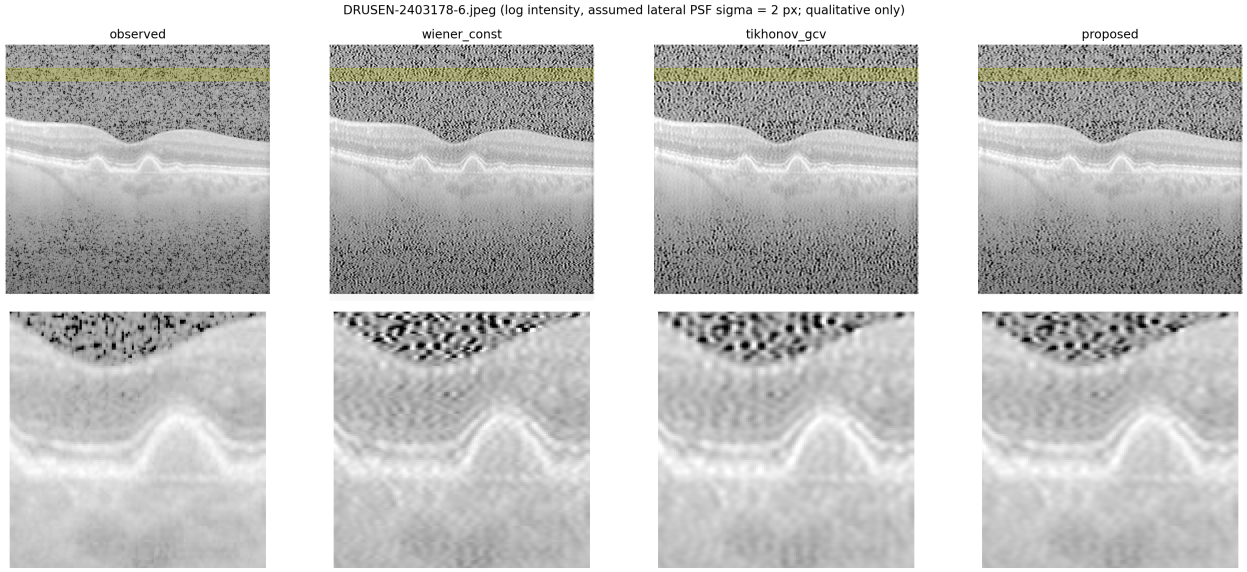


Fig. 12. Qualitative OCT example: observed log-intensity B-scan and lateral deconvolution outputs, with the vitreous background band indicated.

8 dB (Section 9.2). Mirror padding removes most of this; tapering or apodization are alternatives.

C. Limitations

1. Known, shift-invariant PSF. All synthetic results use the true PSF. All real results assume a Gaussian whose width is swept, not measured. Resolution figures on real data are therefore relative, and recovered fine structure is not claimed to be physically real.

2. Small, correlated test set. The synthetic test uses 8 crops from 2 fields of view. Crops of one field are not independent, so the Wilcoxon p-values ($\approx 10^{-80}$) overstate the evidence;

the per-setting means and spreads are the reliable result. The real-noise evaluation uses one raw frame of 2 fields.

3. Reference noise. The FMD 50-frame average still contains about $1/\sqrt{50}$ of the single-frame noise. The real-data noise gain therefore slightly mixes in filtered reference noise; the error is about $1/50$ in variance and does not affect the comparison between methods.

4. OCT is qualitative. Log-speckle is non-Gaussian and correlated, the PSF is assumed, and there is no reference image. Only 13 B-scans were usable after removing archive metadata files.

5. Global baseline settings. Constant Wiener uses one NSR and Richardson–Lucy uses one iteration count across all PSF widths and noise levels. This global-setting constraint creates a compromise: a setting selected by averaging across conditions can be suboptimal for an individual PSF/noise pair. RL was tuned on the development images only; its tuned setting (one iteration, the lowest value of the grid) makes it behave as a smoother (Section 5.1), so it is a weak sharpening baseline, and a stronger RL (for example mirror-padded and initialised from the observed image) was not evaluated. The constant-Wiener NSR of 0.0316 is within its search grid. Per-condition tuning would strengthen these baselines but would no longer represent one globally fixed setting.

D. Future work

- A spatially adaptive λ from local noise estimates, for Poisson–Gaussian data.
- Anisotropic (orientation-binned) spectral fits.
- Joint estimation of the PSF width from the same spectrum, using the location of the noise floor.
- Comparison with total-variation regularization at matched noise gain.

Across the supplied synthetic Gaussian benchmark, the proposed filter approaches the oracle Wiener result and improves on the globally tuned constant-NSR baseline. Its PSNR is effectively tied with Tikhonov–GCV under matched white Gaussian noise. Under Poisson–Gaussian noise, the median paired comparison gives Tikhonov–GCV a small advantage of 0.015 dB, while the proposed method retains gains over constant Wiener, Richardson–Lucy, and unsupervised Wiener. The most defensible practical claim is therefore adaptive regularization with a robust noise-gain profile on the examined real data, not universal superiority in reconstruction accuracy.

The principal limitations remain: a global stationary-noise estimate; an isotropic radial power-law model that cannot represent oriented cellular structures; a known, shift-invariant PSF assumption in synthetic tests and an assumed Gaussian PSF on real images; a small, correlated synthetic test set; residual noise in the 50-frame average; and qualitative-only OCT evidence. PSF mismatch and boundary effects can materially affect performance.

XII. Conclusion

We presented a Wiener-type deconvolution filter whose frequency-dependent regularizer is estimated from the blurred image itself. The noise power comes from the band the optics cannot pass, and the signal spectrum from a noise-corrected power-law fit. The filter costs a few FFTs and has no regularization parameter to tune.

On a controlled benchmark built from real fluorescence-microscopy images:

- it came within 0.15–0.39 dB of the oracle Wiener filter;
- it outperformed a tuned constant-NSR Wiener filter (median +0.72 dB, up to +3.6 dB), Richardson–Lucy (+3.5 dB) and unsupervised Wiener (+3.1 dB);
- it was statistically equivalent to Tikhonov regularization with GCV.

The main limitations are a global noise model, an isotropic signal model and an assumed PSF on real data. These point to spatially adaptive and anisotropic extensions.

In summary, the proposed method estimates both noise power and a parametric signal spectrum from the observed image, yielding a frequency-dependent Wiener regularizer without per-image manual tuning. The updated evidence supports performance close to the oracle and strong gains over a single globally tuned NSR, while showing practical equivalence to Tikhonov–GCV on Gaussian synthetic PSNR and a small Tikhonov–GCV advantage under Poisson–Gaussian noise.

XIII. Reproducibility Appendix

Environment used is a local CPU machine running Jupyter, with the repository folder as the working directory (or the folder named by the environment variable ‘DSP_DECONV_ROOT‘). Every notebook starts with the same setup cells: paths, packages, config. Package versions are in ‘results/versions.txt‘.

Folder layout: noise-aware-deconvolution/

- lib/ forward.py data.py metrics.py baselines.py proposed.py
- configs/ params.yaml (grids, seeds) tuned.yaml (dev-tuned NSR, RL iterations) split.yaml (dev/test)
- data/raw/ FMD/<subset>/{raw,gt}/<fov>/*.png OCT/*.jpeg
- results/ *.csv tables/*.csv versions.txt figures/ *.png

The notebooks must be run in the order 00, 01, 03, 02, 04, 05, because 02 reads the results written by 03. The ‘lib/*.py‘ files are plain source files imported by the notebooks; edits take effect through the ‘autoreload‘ extension.

All noise is drawn with ‘numpy.random.default_rng(seed)‘, seeds 0–4, and ‘unsupervised_wiener‘ is called with ‘rng=0‘. Re-running the notebooks therefore reproduces the CSVs exactly, as long as the package versions match.

The environment record of the final run lists Python 3.12.9, NumPy 2.5.4, SciPy 1.18.1, and scikit-image 0.26.0. The six notebooks supplied for the project are 00_setup_data, 01_forward_baselines_new, 02_proposed_new, 03_experiments_new, 04_real_data_new, and 05_figures_new. The run order should be followed as documented in the README and this appendix.

Table 22: Notebooks of the reproducibility package, the cells they run, and the files they produce.

Notebook	Cells	Produces
00_setup_data	all	downloads FMD subsets and the OCT sample
01_forward_baselines_new	all	forward-model and baseline checks; tunes the baselines and writes ‘tuned.yaml‘
03_experiments_new	all	‘synthetic_gauss_test.csv‘, ‘synthetic_pg_test.csv‘, ‘edge_test.csv‘, ‘wilcoxon.csv‘, ‘psf_mismatch.csv‘, ‘boundary.csv‘
02_proposed_new	all (after 03)	‘ablation.csv‘
04_real_data_new	all	‘split.yaml‘, ‘versions.txt‘, ‘real_fmd.csv‘, ‘real_oct.csv‘, real-data figures
05_figures_new	all	‘fig1‘–‘fig7‘, ‘results/tables/table*.csv‘

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